Comparison of Equilibration Approaches for PVC/DEHP System

(16 PVC-20 + 34 DEHP, ~40 wt% DEHP)

System Details (A)

- Composition: 16 PVC 20-mers + 34 DEHP
- Force field: OPLS-AA (PolyParGen), fixed the atom naming discrepancies in the itp files and total system is charge has been carefully optimized to 0 instead of the minor charges generated by ligpargen, since PME requires the system to be at 0 (integer)
- No warnings in energy minimization and npt, system has a net charge of 0
- Barostat and Thermostat: Berendsen and V-rescale
- Simulation details: 50 ns with 6K/ns cooling rate from 600K to 300K

Method

Protocol

- High-T equilibration at 600 K
- Controlled cooling 600 → 300 K at 6 K/ns (50 ns) under NPT
- Further NPT at 300 K to find density fluctuations

Current / Final Results

Quantity	**Value**
Density at ~600K	**~0.90-0.91 g/cm^3**
Density at ~500K	**~**
Density at ~400K	
Density at ~300K	**~1.19-1.207 g/cm^3**
Box size	**~3.60 nm at 600K and 3.88 nm at 300K**
Cooling rate	**6 K/ns**

Observations

- …

System Details (B)

- Composition: 16 PVC 20-mers + 34 DEHP
- Force field: OPLS-AA (PolyParGen)
- Barostat and Thermostat: Berendsen and V-rescale
- Simulation details: 5ns npt at 600K and then an 8ns npt at 300K (3 times)

Results

Quantity	Value
Density at 600K	**~0.9 g/cm^3**
Density at 300K	**~1.75-1.8 g/cm^3**

Observations

- …

System Details (C)

- Composition: 12 PVC 20-mers + 13 DEHP
- Force field: OPLS-AA (PolyParGen)
- Barostat and Thermostat: Berendsen and V-rescale
- Simulation details: 5ns npt at 600K and then an 8ns npt at 300K (3 times)

Results

Density at 600K	**~0.8 g/cm^3**
Density at 300K	**~1.23 g/cm^3**

Observations

- …

System Details (D)

- Composition: 16 PVC 20-mers + 34 DEHP
- Force field: OPLS-AA (Vanilla) and OPLS-AA charges
- Barostat and Thermostat: C-rescale and V-rescale
- Simulation details: 0.5 ns nvt equil (to generate velocities) at 600K and then an 10ns npt equil at 600K and then temperature cyclings (2-3)

Results

Density at 600K	**0.94 g/cm^3**
Density at 300K	**1.18-1.19 gm/cm^3**

Observations

- …

Results:

MSD (200 ps), DEHP and PVC centre of mass

T (K)	DEHP	PVC
594	0.864	0.452
504	0.259	0.131
444	0.085	0.039
~400	0.026	0.010
~350	0.005	0.003
~305–310	0.004	0.002

MSD-based Tg**:** DEHP ≈ 348 K (337–352 K), PVC ≈ 351 K (350–375 K). They freeze together.

DEHP is about 1.6–2.7× more mobile than PVC at every temperature.

Diffusion coefficient, DEHP (10^{-5} cm²/s; only above ~440 K, where the MSD becomes diffusive)

T (K)	594	564	534	504	474	444
D	0.866	0.542	0.507	0.215	0.123	0.063

- Arrhenius fit: **Ea ≈ 39 kJ/mol** (R^2 = 0.98).
- No PVC D: the chains don't reach the diffusive regime in 2 ns windows.

Ree and Rg, PVC 20-mer

	Ree (nm)	**Rg (nm)**
Melt (~600 K)	~2.4–2.5	~0.93–0.97
Final (300 K)	2.88 ± 0.65 (chain spread; ±0.16 on the mean)	0.99 ± 0.10
Paper (Fig. 1)	~2.9	~1.1

- Chain shape stops changing below ~420 K.
- Largest Ree seen: 4.57 nm, larger than the 3.6 nm box meaning that the PVC was interacting with its own periodic image frequently